Polisetti Kailash Bharat

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.97
Sri Chaitanya Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh, India)	2024	96.00%
Sri Chaitanya E.M. School	Class X (Board of Secondary Education, Andhra Pradesh, India)	2022	91.83%

SCHOLASTIC ACHIEVEMENTS

Achieved highest branch **SGPA** of **9.5** in 2nd semester

Ranked among the **top three** students in branch based on cumulative GPA

Achieved **Codeforces Expert** rating of **1602**, demonstrating strong analytical thinking and problem solving skills.

Earned **(10/10)** grade in courses: Differential Equations, Probability, Calculus II, and Computer Aided Numerical Methods II

Awarded **Gold Medals** in **International Mathematics Olympiad** and **National Science Olympiad**

PROJECTS

Health Centre Website:

- Built using Flask, MySQL, HTML, and CSS to manage and display **real time medicine stock**.
- Implemented search with **live suggestions**, designed clear **stock indicators**. Developed **secure staff login** and update system for managing stock changes.
- **Solved a real campus problem**: Students don't need to visit the health centre just to check medicine availability.

Finite Difference Schemes for Linear Advection Equation

- Implemented and compared **Numerical schemes** (FTBS, FTFS, FTCS, Lax Friedrichs, Lax Wendroff, Crank Nicolson) for solving the **1D Linear Advection Equation**.
- Optimized algorithms for **performance and accuracy**, handling stability constraints.
- Demonstrated strong skills in **numerical methods, algorithm design, and cross language implementation.**

xv6 Operating System extensions:

Course Project, IIT Hyderabad

- Engineered **custom kernel system calls and process statistics** (PID/PPID retrieval, child process tracking, syscall counters) with proper locking and integration into xv6's syscall interface.
- Designed and implemented a **Multi Level Feedback Queue scheduler** and **Clock based demand paging** with in memory swap, lazy allocation, and per process metrics, ensuring stability under memory pressure.
- Built **disk swap subsystem with RAID (0/1/5) and disk scheduling (FCFS, SSTF).**

Molecular Dynamics Engine

Course Project, IIT Hyderabad

- Built a **Molecular Dynamics simulation engine in Python** implementing **LJ Potential, Periodic Boundary**

Conditions, Velocity verlet integrator, with shifted force correction for computational efficiency.
- Designed and validated via **Radial Distribution Function**. Useful in **Materials science** studying phase transitions, atomic-scale behavior.

SKILLS

Languages: C, C++, Python

Libraries: Numpy, Scipy, Pandas, Matplotlib, Flask

Web Development: HTML, CSS (Databases: MySQL)

Tools & Software: Git/GitHub, Visual Studio Code, Jupyter Notebook, Quantum Expresso

Soft Skills: Leadership, Adaptability, Teamwork, Event Execution, Communication, Dedication

RELEVANT COURSES

Computer Science courses: Programming, Data Structures and its Applications, Operating Systems

Math courses: Probability, Linear Algebra, Calculus I & II, Differential Equations, Statistics, Complex Variables, Transform Techniques

Computational Courses: Fundamentals of Scientific Computing, Computer Aided Numerical Methods - 1&2, Computational Methods in Material Science, Electronic Structure and Atomistic Modeling of Materials

POSITIONS OF RESPONSIBILITY

Class representative of Computational Engineering (2025-2026):

- Actively liased between students and administration, ensuring concerns were addressed promptly.
- Successfully launched the first official department website of Computational Engineering.
- Managed 50K+ budget for Departmental Day as Class Representative, ensuring efficient allocation and execution.

Head Internship Coordinator of Computational Engineering (2026-2027):

- Developed a structured brochure on the branch, facilitating Industry engagement and Internship opportunities.
- Led internship coordination, managing student-recruiter communication to streamline Internships.

Logistics Coordinator of 10th Intercon Spic Macay - Largest cultural immersion program hosted by IITH

- Coordinated logistics for 1,400+ participants and Padma Shri awardees, ensuring smooth accommodation.
- Managed end to end event operations, covering venue setup, resource allocation, and daily activity tracking.

EXTRACURRICULAR

- Achieved a commendable CGPA of **9.00** while simultaneously holding **9 Leadership positions** across diverse student bodies at IIT Hyderabad, showcasing strong multitasking ability, and effective people management skills.
- Ranked in the top **0.1%** globally on Matiks, with a rating of **2391** in memory puzzles
- Contributed **100+ hours** of community service through active participation in NSS events
- Trained in swimming at IIT Hyderabad, enhancing fitness and discipline alongside academics.